

Volume I · Chapter 2 — Foundations of Clinical Data (EHR; code sets; privacy) · Labs

Source: split from med-tech-curriculum-V1.md (V1). From this split onward this chapter is maintained **independently** here. See Volume-I-Chapter-2-context.md for the AI interaction log.

Prerequisites: Ch 1 (Python/Pandas).

Learning outcomes — the student can:

- Describe **EHR structure** and its core data domains, and distinguish structured vs. unstructured data.
- Navigate and **map medical code sets** — ICD-10-CM/PCS, CPT/HCPCS, SNOMED CT, LOINC — to the right purpose.
- Explain **data privacy/regulation** (HIPAA, GDPR) and what makes clinical data identifiable.
- Apply basic **de-identification** and justify **synthetic-data** use for development.

Labs (hands-on) — 6 h:

#	Lab	h
2a	Explore an EHR schema; classify fields into data domains; structured vs. unstructured	2
2b	Code lookup & mapping: build a small ICD-10 ↔ SNOMED ↔ LOINC crosswalk in Pandas	2
2c	De-identification: apply HIPAA Safe Harbor (18 identifiers) to a synthetic dataset; verify	2

Datasets/tools: synthetic EHR; public browsers (ICD-10, SNOMED CT browser, LOINC search); Pandas.

Assessment: quiz on code sets & privacy (**40%**); crosswalk deliverable (**30%**); de-identification lab rubric (**30%**).

Key decisions: which terminology fits which purpose; when data is/ isn't de-identified; **synthetic vs. real** data for development.

References: plan §13 → *Clinical data, code sets & privacy*.

Hours: Theory **10** + Lab **6** = **16**.